Neosha Gupta Narayanan

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Education

Georgia Institute of Technology — **Atlanta, GA**
PhD Candidate in Glacier Geophysics; Advisor: Winnie Chu — *August 2023-Present*

Massachusetts Institute of Technology (MIT) — **Cambridge, MA**
Master of Science in Geophysics and Seismology; Advisor: Brent Minchew — *July 2023*
Bachelor of Science in Materials Science and Engineering — *May 2022*

Amherst Regional High School — **Amherst, MA**
Valedictorian — *June 2018*

Publications

- **Narayanan, N**., Sommers, A., Chu, W., Steiner, J., Siddique, A.M., Meyer, C., Minchew, B. (2025). Simulating Seasonal Evolution of Subglacial Hydrology at a Surging Glacier in the Karakoram. *Journal of Glaciology.* doi.org/10.1017/jog.2025.10078
- **Narayanan, N.** (2023). *Modeling Subglacial Hydrology in the Himalayas*. Master's thesis, Massachusetts Institute of Technology. https://hdl.handle.net/1721.1/153347
- Su, I., **Narayanan, N.,** et al. In-situ spider web construction and mechanics. *Proceedings of the National Academy of Sciences (PNAS),* August 2021.
- Su, I., Jung, G.S., **Narayanan, N.,** Buehler, M. Perspectives on 3D printing of self-assembling materials and structures. *Current Opinion in Biomedical Engineering*, February 2020.
- Angles 2019 (MIT Writing Journal): "Homeland", "Saving our Pollinators" (not peer-reviewed)

Conference Proceedings

- **Narayanan, N.,** Chu, W., Sommers, A., Meyer, C., Steiner, J., Minchew, B. (2026, April). *Investigating Relationships between Lake Drainages and Surge Motion through Modeling in the Karakoram Himalaya*. European Geosciences Union General Assembly (POSTER).
- **Narayanan, N.,** Chu, W., Sommers, A., Meyer, C., Steiner, J., Robel, A., Minchew, B. (2025, December). *Coupled Subglacial Hydrology and Ice Dynamics Modeling of Lake Drainages and Surging Glaciers in the Himalayas*. American Geophysical Union Fall Meeting (ORAL PRESENTATION).
- Ishraque, F., Afzal Shadab, M., Lauter, O., **Narayanan, N.,** et al. (2025, December). *Assessing Challenges for Polar Early Career Scientists During Science Policy Upheaval* (invited). American Geophysical Union Fall Meeting.
- **Narayanan, N.,** Chu, W., Meyer, C., Poinar, K., Sommers, A., Mejia, J., et al. (2024, December). *Radar Measurements of Firn Aquifer at Helheim Glacier, Greenland*. American Geophysical Union Fall Meeting (POSTER).
- Poinar, K., Sommers, A., Mejia, J., Meyer, C., **Narayanan, N**., et al. (2024, December). *Hydrologic Implications of the Water Feeding Ice Caves at Helheim Glacier, Greenland*. American Geophysical Union Fall Meeting (POSTER).
- Mejia, J., Poinar, K., Meyer, C., Sommers, A., **Narayanan, N.**, Chu, W. (2024, December). *Direct Observations of Mixed-Mode Fracture Opening of a Water-Fed Crevasse on Helheim Glacier, Greenland.* Amerian Geophysical Union Fall Meeting (POSTER).
- **Narayanan, N.,** Sommers, A., Chu, W., Steiner, J., Siddique, A.M., Meyer, C., Minchew, B. (2024, April*). Coupled Ice Dynamics and Subglacial Hydrology Simulations of Glacier Surges in the Himalayas*. European Geosciences Union (POSTER). https://doi.org/10.5194/egusphere-egu24-11819
- **Narayanan, N.,** Sommers, A., Steiner, J., Siddique, A.M., Minchew, B. (2023, December). *Modeling Subglacial Hydrology in the Himalayas: Applications to Surges and Glacial Lake Outburst Floods.* American Geophysical Union Fall Meeting. (POSTER)
- **Narayanan, N.,** Millstein, J., & Minchew, B. (2022, December). *Simulation and Analysis of Deformation and Stability in Antarctic Ice Shelves*. American Geophysical Union Fall Meeting. (POSTER)
- **Narayanan, N.,** Millstein, J., & Minchew, B. (2022, April). *Simulation and Analysis of Deformation in Antarctic Ice Shelves*. Northeast Glaciology Meeting. (POSTER)

Awards, Scholarships, and Fellowships

Dartmouth College - Evans Family Fellow — *2025*

Polar Impact Mentorship Initiative (PIMI) Mentee Cohort 2025-26	*2025-26*
Georgia Tech Scheller College Graduate Sustainability Fellow	*2023-24*
MIT - William Asbjornsen Albert Memorial Fellow	*2022-23*
MIT Priscilla King Gray (PKG) Summer Fellow	*2021*
MIT Burchard Scholar	*2020*
Henry David Thoreau Scholar	*2018-2022*

Invited Talks

- Maths on Ice Seminar: "Coupled Subglacial Hydrology and Ice Dynamics Modeling of Lake Drainages & Glacier Motion in the Himalayas." December 11, 2025.
- "The Cool Science of Glaciers! With Neosha Narayanan" Spectacular Science Podcast with Akshay Jayaraman. October 2024. https://www.youtube.com/watch?v=VQ9Afvq5NHk
- Georgia Tech Geophysics Department Seminar**,** April 26, 2024
- Keynote speaker, 2022 Henry David Thoreau Society Annual Meeting
- Invited panelist, MIT South Asian Alumni Association's International Women's Day Panel (Spring 2023)
- Student moderator, MIT "Intuitively Obvious" video series
    - Moderated filmed conversations with AAPI students as part of the 2022 "Intuitively Obvious" video series on the manifestations of race and privilege at MIT

Outreach & Professional Activities

Manuscript referee, npj Natural Hazards and Scientific Advances (Nature)	*2026-Present*
Manuscript referee, The Cryosphere (Copernicus Journals)	*2025-Present*
Co-Chair, US Association of Polar Early Career Scientists (USAPECS)	*Oct 2025-Present*
Vice President, Georgia Tech Graduates in Earth and Atmospheric Sciences (GEAS)	*August 2025-Present*
Georgia Tech Clough Climate Seminar Committee	*2025-2026*
Georgia Tech EAS Graduate Research Symposium Chair	*2025*
USAPECS Executive Board Member	*2024-2025*
Letters to a Pre-Scientist STEM Professional Volunteer	*2023-2025*
MIT Admissions Alumni Educational Counselor (EC)	*2022-Present*
MIT South Asian Association of Students (SAAS) President	*2021- 22*
MIT Cycling Club Recruitment Officer	*2020-2021*

Professional Memberships & Affiliations

American Geophysical Union	*2022-Present*
European Geophysical Union	*2024-Present*
International Glaciological Society	*2024-Present*

Field Experience

- May 2026: Greenland Ice-Sheet and Ocean Summer School participant in Nuuk, Greenland
- July 2025: "Follow the Water: Hydrology at Helheim Glacier" (funded by the Heising-Simons Foundation) – deployed radar echo-sounders (ApRES), ground-penetrating radar (PulseEKKO and UAF Groundhog), and carried out common offset, common midpoint, and multi-offset surveys. Successfully carried out retrieval of subsurface UNAVCO GPS units. Part of a collaborative expedition on the Greenland Ice Sheet.
- July 2024: "Follow the Water: Hydrology at Helheim Glacier" – deployed ApRES and seismo-electric instruments, dug snow pits, and assisted with equipment retrieval
- June 2024: Participated in the International Summer School in Glaciology at McCarthy, Alaska hosted by University of Alaska Fairbanks
- June 2023: MIT-EAPS Crosby Trip: Learned about geology and sedimentology during a field expedition in Sicily, Italy
- July 2022: With IIT Roorkee scientists, conducted a field study on Suraj Tal in Himachal Pradesh, India
- January 2022: MIT-CEE Traveling Research Experience (TREX) - conducted fieldwork on invasive plant species distribution and air quality on the island of Hawaii
- August 2019: MIT-EAPS Yellowstone First-Year Orientation Program Teaching Assistant: taught incoming undergraduate students about geology in Yellowstone National Park

Teaching and Mentoring

GT EAS 2600 (Earth Processes) - Laboratory Teaching Assistant *2023*

- Wrote and delivered lectures on introductory earth science topics, guided students through hands-on earth science lab assignments, held office hours to assist with conceptual questions, and graded lab assignments.

MIT Terrascope: Undergraduate Teaching Fellow (UTF) *2020-2022*

- Mentored students, held office hours, and ran meetings for fall Solving Complex Problems class which focuses on a different sustainability or climate related issue each year
- Evaluated and provided feedback on student-produced website and presentation drafts prior to final presentation
- Collaborated with other UTFs to research & develop innovative methods for remote collaborative project work during COVID-19 pandemic

MIT Associate Advising Program *2019-2022*

- Assisted faculty advisors with advising first-year students at MIT by providing advice on class registration and helping students find resources for academic performance and mental health
- Mentored 21 total first-year students both in-person and during the pandemic through weekly one-on-one meetings

Research Experience

Indian Institute of Technology, Roorkee Hydrology Department **Roorkee, Uttarakhand, India**
Guide: Prof. Dhyan Singh Arya *June-August 2022*

MIT EAPS Glacier Dynamics & Remote Sensing Group **Cambridge, MA (Remote)**
Advisor: Joanna Millstein; PI: Brent Minchew *May 2020-May 2021*

MIT Laboratory for Atomistic and Molecular Materials **Cambridge, MA**
Advisor & PI: Markus Buehler *Jan 2019-June 2020*

MIT EAPS Susan Solomon Lab **Cambridge, MA**
Undergraduate Researcher *Sep-Dec 2018*

Work Experience

Charles River Watershed Association **Cambridge, MA**
Data Analysis Intern *June-August 2021*

MIT Angles Magazine **Cambridge, MA**
Editorial Assistant *June 2020-August 2021*

MIT-South Asia Oral History Project **Cambridge, MA (remote)**
Student Intern, MIT History Department *January 2020, 2021*